

Template

Studentnames and studentnumbers here

2026-06-12

Set-up your environment

```
library(cbsodataR)
library(dplyr)
library(tidyr)
library(sf)
library(knitr)
library(ggplot2)
```

Title Page

Include your names Melvin Hofman Max Bergsma Gijs Stevens Hidde Voors Imad el Hilali

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Housing affordability has become an increasingly important social issue in the Netherlands over the past decade. House prices have risen substantially in many parts of the country, while income growth may not have kept pace. As a result, many households, particularly first-time buyers and younger generations, face greater difficulties entering the housing market. Rising housing costs can contribute to financial stress, wealth inequality, and reduced access to homeownership.

Understanding whether house prices have increased faster than incomes is therefore important for evaluating changes in housing affordability. If housing prices grow more rapidly than household incomes, purchasing a home becomes increasingly difficult for a large share of the population. This may have significant social and economic consequences, including increased inequality between homeowners and non-homeowners.

The main research question of this study is:

Have housing prices in the Netherlands increased faster than incomes over the past ten years?

To answer this question, the following sub-questions are examined: 1. How have house prices and average incomes developed between 2014 and 2024? 2. How has housing affordability changed during this period? 3. Are there significant differences in housing affordability between Dutch provinces? 4. Did the COVID-19 pandemic influence the relationship between house prices and income?

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
# data_download <- cbs_get_data("70072NED")

# Add region label to the dataset
# raw_dataset <- data_download %>% cbs_add_label_columns()
# write.csv(raw_dataset, "data/raw_dataset.csv")

# Load saved dataset
raw_dataset = read.csv("data/raw_dataset.csv")
```

2.2 Provide a short summary of the dataset(s)

```
huizenprijzen_provincie = raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year == 2023,
    startsWith(RegioS, "PV")
  ) %>%
  select(
    Year,
    RegioS,
    RegioS_label,
    GemiddeldeWOZWaardeVanWoningen_98
  )
write.csv(huizenprijzen_provincie, "data/huizenprijzen_provincie.csv")

kable(huizenprijzen_provincie)
```

Year	RegioS	RegioS_label	GemiddeldeWOZWaardeVanWoningen_98
2023	PV20	Groningen (PV)	268
2023	PV21	Fryslân (PV)	277
2023	PV22	Drenthe (PV)	294
2023	PV23	Overijssel (PV)	325
2023	PV24	Flevoland (PV)	347
2023	PV25	Gelderland (PV)	364
2023	PV26	Utrecht (PV)	447
2023	PV27	Noord-Holland (PV)	460
2023	PV28	Zuid-Holland (PV)	359
2023	PV29	Zeeland (PV)	281
2023	PV30	Noord-Brabant (PV)	370
2023	PV31	Limburg (PV)	288

```
inkomen_provincie <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
```

```

) %>%
filter(
  Year == 2023,
  startsWith(RegioS, "PV")
) %>%
select(
  Year,
  RegioS,
  RegioS_label,
  BronInkomenAlsWerknemer_141
)
write.csv(inkomen_provincie, "data/inkomen_provincie.csv")

kable(inkomen_provincie)

```

Year	RegioS	RegioS_label	BronInkomenAlsWerknemer_141
2023	PV20	Groningen (PV)	37.5
2023	PV21	Fryslân (PV)	37.9
2023	PV22	Drenthe (PV)	39.7
2023	PV23	Overijssel (PV)	39.5
2023	PV24	Flevoland (PV)	40.4
2023	PV25	Gelderland (PV)	40.6
2023	PV26	Utrecht (PV)	44.4
2023	PV27	Noord-Holland (PV)	44.7
2023	PV28	Zuid-Holland (PV)	42.4
2023	PV29	Zeeland (PV)	40.3
2023	PV30	Noord-Brabant (PV)	41.8
2023	PV31	Limburg (PV)	39.9

```

provincie_samengevoegd = huizenprijzen_provincie %>%
  inner_join(
    inkomen_provincie,
    by = c("Year", "RegioS", "RegioS_label")
  ) %>%
  rename(
    GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
    InkomenWerknemer = BronInkomenAlsWerknemer_141
  ) %>%
  mutate(statcode = RegioS)

kable(provincie_samengevoegd, digits = 2)

```

Year	RegioS	RegioS_label	GemiddeldeWOZ	InkomenWerknemer	statcode
2023	PV20	Groningen (PV)	268	37.5	PV20
2023	PV21	Fryslân (PV)	277	37.9	PV21
2023	PV22	Drenthe (PV)	294	39.7	PV22
2023	PV23	Overijssel (PV)	325	39.5	PV23
2023	PV24	Flevoland (PV)	347	40.4	PV24
2023	PV25	Gelderland (PV)	364	40.6	PV25
2023	PV26	Utrecht (PV)	447	44.4	PV26
2023	PV27	Noord-Holland (PV)	460	44.7	PV27
2023	PV28	Zuid-Holland (PV)	359	42.4	PV28

Year	RegioS	RegioS_label	GemiddeldeWOZ	InkomenWerknemer	statcode
2023	PV29	Zeeland (PV)	281	40.3	PV29
2023	PV30	Noord-Brabant (PV)	370	41.8	PV30
2023	PV31	Limburg (PV)	288	39.9	PV31

```
landelijk <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year >= 2015,
    Year <= 2024,
    startsWith(RegioS, "NL")
  ) %>%
  select(
    Year,
    RegioS_label,
    BronInkomenAlsWerknemer_141,
    GemiddeldeWOZWaardeVanWoningen_98
  ) %>%
  rename(
    GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
    InkomenWerknemer = BronInkomenAlsWerknemer_141
  )

write.csv(landelijk, "data/landelijk.csv")

kable(landelijk)
```

Year	RegioS_label	InkomenWerknemer	GemiddeldeWOZ
2015	Nederland	30.3	206
2016	Nederland	31.4	208
2017	Nederland	32.0	217
2018	Nederland	32.3	230
2019	Nederland	33.7	250
2020	Nederland	35.3	271
2021	Nederland	37.1	290
2022	Nederland	39.0	317
2023	Nederland	41.8	368
2024	Nederland	44.7	378

```
provincie_summary = provincie_samengevoegd %>%
  summarise(
    Mean_WOZ = mean(GemiddeldeWOZ),
    Min_WOZ = min(GemiddeldeWOZ),
    Max_WOZ = max(GemiddeldeWOZ),
    SD_WOZ = sd(GemiddeldeWOZ),

    Mean_Income = mean(InkomenWerknemer),
    Min_Income = min(InkomenWerknemer),
    Max_Income = max(InkomenWerknemer),
```

```

    SD_Income = sd(InkomenWerknemer),
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = c("Statistic", "Variable"),
    names_sep = "_",
    values_to = "Value"
  ) %>%
  pivot_wider(
    names_from = Statistic,
    values_from = Value
  ) %>%
  mutate(
    Variable = recode(
      Variable,
      WOZ = "Average WOZ Value",
      Income = "Employee Income",
      Affordability = "WOZ-to-Income Ratio"
    )
  ) %>%
  select(Variable, Mean, Min, Max, SD)

kable(
  provincie_summary,
  caption = "Descriptive statistics for home value in provinces in the Netherlands, 2023",
  digits = 2
)

```

Table 5: Descriptive statistics for home value in provinces in the Netherlands, 2023

Variable	Mean	Min	Max	SD
Average WOZ Value	340.00	268.0	460.0	64.25
Employee Income	40.76	37.5	44.7	2.24

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

```
provincie_samengevoegd <- provincie_samengevoegd %>%
  mutate(
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
  )
provincie_samengevoegd
```

```
##   Year RegioS      RegioS_label GemiddeldeWOZ InkomenWerknemer statcode
## 1  2023 PV20      Groningen (PV)          268           37.5     PV20
## 2  2023 PV21      Fryslân (PV)          277           37.9     PV21
## 3  2023 PV22      Drenthe (PV)          294           39.7     PV22
## 4  2023 PV23      Overijssel (PV)        325           39.5     PV23
## 5  2023 PV24      Flevoland (PV)         347           40.4     PV24
## 6  2023 PV25      Gelderland (PV)        364           40.6     PV25
## 7  2023 PV26      Utrecht (PV)          447           44.4     PV26
## 8  2023 PV27      Noord-Holland (PV)      460           44.7     PV27
## 9  2023 PV28      Zuid-Holland (PV)      359           42.4     PV28
## 10 2023 PV29      Zeeland (PV)           281           40.3     PV29
## 11 2023 PV30      Noord-Brabant (PV)      370           41.8     PV30
## 12 2023 PV31      Limburg (PV)          288           39.9     PV31
```

```
##   AffordabilityRatio
```

```
## 1      7.146667
## 2      7.308707
## 3      7.405542
## 4      8.227848
## 5      8.589109
## 6      8.965517
## 7     10.067568
## 8     10.290828
## 9      8.466981
## 10     6.972705
## 11     8.851675
## 12     7.218045
```

```
write.csv(provincie_samengevoegd, "data/provincie_samengevoegd.csv")
```

```
landelijk = landelijk %>%
  mutate(
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer,
    WOZ_growth_yoy = (GemiddeldeWOZ / lag(GemiddeldeWOZ) - 1) * 100,
    Income_growth_yoy = (InkomenWerknemer / lag(InkomenWerknemer) - 1) * 100,
    GrowthGap_yoy = WOZ_growth_yoy - Income_growth_yoy
  )
kable(landelijk)
```

Year	RegioS_label	InkomenWerknemer	GemiddeldeWOZ	AffordabilityRatio	WOZ_growth_yoy	Income_growth_yoy	GrowthGap_yoy
2015	Nederland	30.3	206	6.798680	NA	NA	NA
2016	Nederland	31.4	208	6.624204	0.9708738	3.630363	-2.659489
2017	Nederland	32.0	217	6.781250	4.3269231	1.910828	2.416095
2018	Nederland	32.3	230	7.120743	5.9907834	0.937500	5.053283
2019	Nederland	33.7	250	7.418398	8.6956522	4.334365	4.361287
2020	Nederland	35.3	271	7.677054	8.4000000	4.747775	3.652226
2021	Nederland	37.1	290	7.816712	7.0110701	5.099150	1.911920

Year	RegioS_label	InkomenWerknemer	GemiddeldeWOZ	AffordabilityRatio	WOZ_growth	Income_growth	GrowthGap_yoy
2022	Nederland	39.0	317	8.128205	9.3103448	5.121294	4.189051
2023	Nederland	41.8	368	8.803828	16.0883281	7.179487	8.908841
2024	Nederland	44.7	378	8.456376	2.7173913	6.937799	-4.220408

3.3 Visualize temporal variation

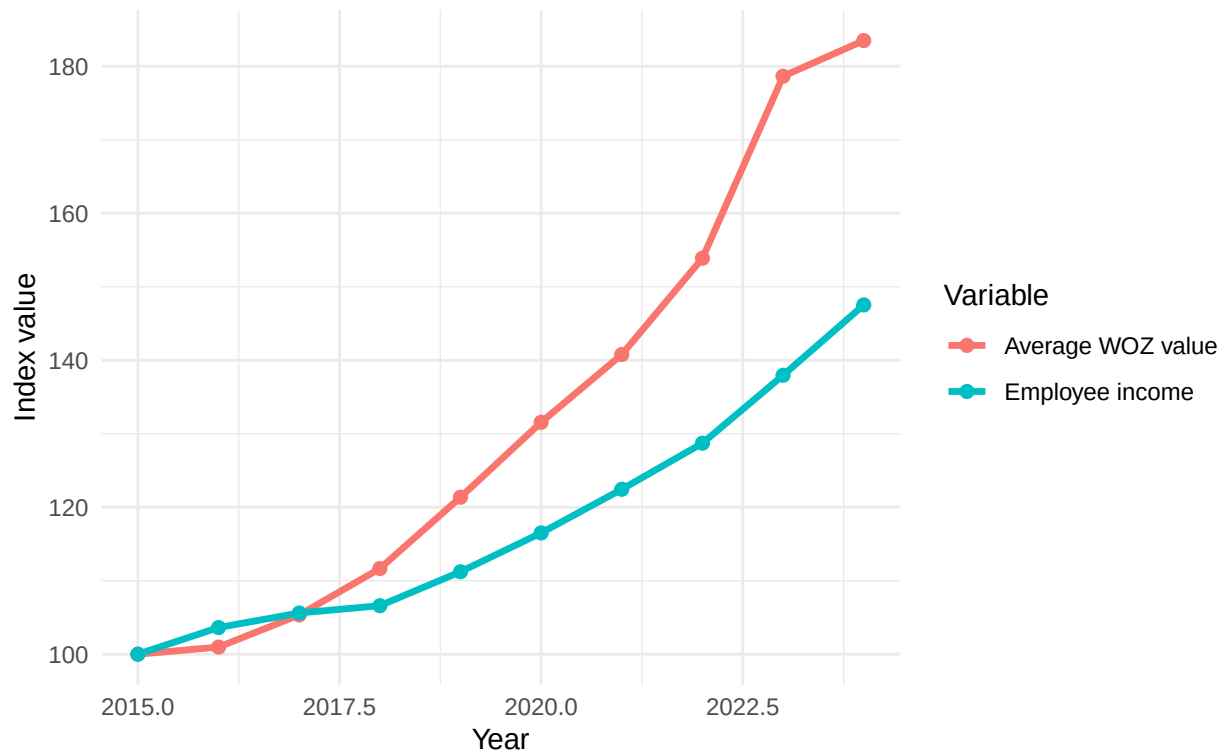
```
# Deze plot gemaakt met chat

landelijk_index <- landelijk %>%
  arrange(Year) %>%
  mutate(
    WOZ_index = GemiddeldeWOZ / GemiddeldeWOZ[Year == 2015] * 100,
    Inkomen_index = InkomenWerknemer / InkomenWerknemer[Year == 2015] * 100
  ) %>%
  select(Year, WOZ_index, Inkomen_index) %>%
  pivot_longer(
    cols = c(WOZ_index, Inkomen_index),
    names_to = "Variable",
    values_to = "Index"
  ) %>%
  mutate(
    Variable = recode(
      Variable,
      WOZ_index = "Average WOZ value",
      Inkomen_index = "Employee income"
    )
  )

ggplot(landelijk_index, aes(x = Year, y = Index, color = Variable)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Development of House Values and Employee Income in the Netherlands",
    subtitle = "Indexed values, 2015 = 100",
    x = "Year",
    y = "Index value",
    color = "Variable"
  ) +
  theme_minimal()
```

Development of House Values and Employee Income in the Netherlands

Indexed values, 2015 = 100



3.4 Visualize spatial variation

Here you provide a description of why the plot above is relevant to your specific social problem.

```
provincie_map <- cbs_get_sf("provincie", 2023, verbose = TRUE)

## Reading layer `provincie_2023' from data source
## `https://cartomap.github.io/nl/rd/provincie_2023.geojson' using driver `GeoJSON'
## Simple feature collection with 12 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: 13565 ymin: 306960 xmax: 277529 ymax: 619172
## Projected CRS: Amersfoort / RD New

provincie_samengevoegd <- provincie_samengevoegd %>%
  mutate(
    statcode = trimws(RegioS),
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
  )

provincie_map <- provincie_map %>%
  mutate(
    statcode = trimws(statcode)
  )

provincie_map_data <- provincie_map %>%
  left_join(
```

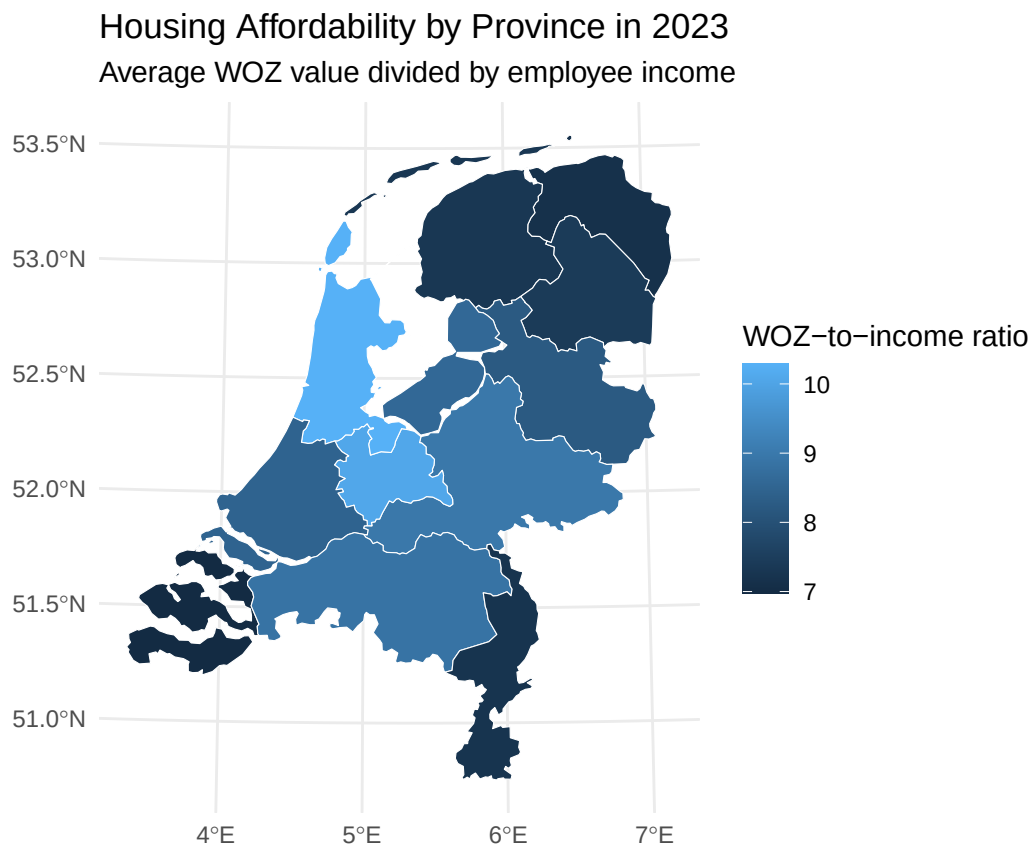


```

    provincie_samengevoegd,
    by = "statcode"
  )

ggplot(provincie_map_data) +
  geom_sf(aes(fill = AffordabilityRatio), color = "white") +
  labs(
    title = "Housing Affordability by Province in 2023",
    subtitle = "Average WOZ value divided by employee income",
    fill = "WOZ-to-income ratio"
  ) +
  theme_minimal()

```



3.5 Visualize sub-population variation

What is the poverty rate by state?

Here you provide a description of why the plot above is relevant to your specific social problem.

```

urban_affordability <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year == 2024,
    startsWith(RegioS, "GM")
  ) %>%
  select(

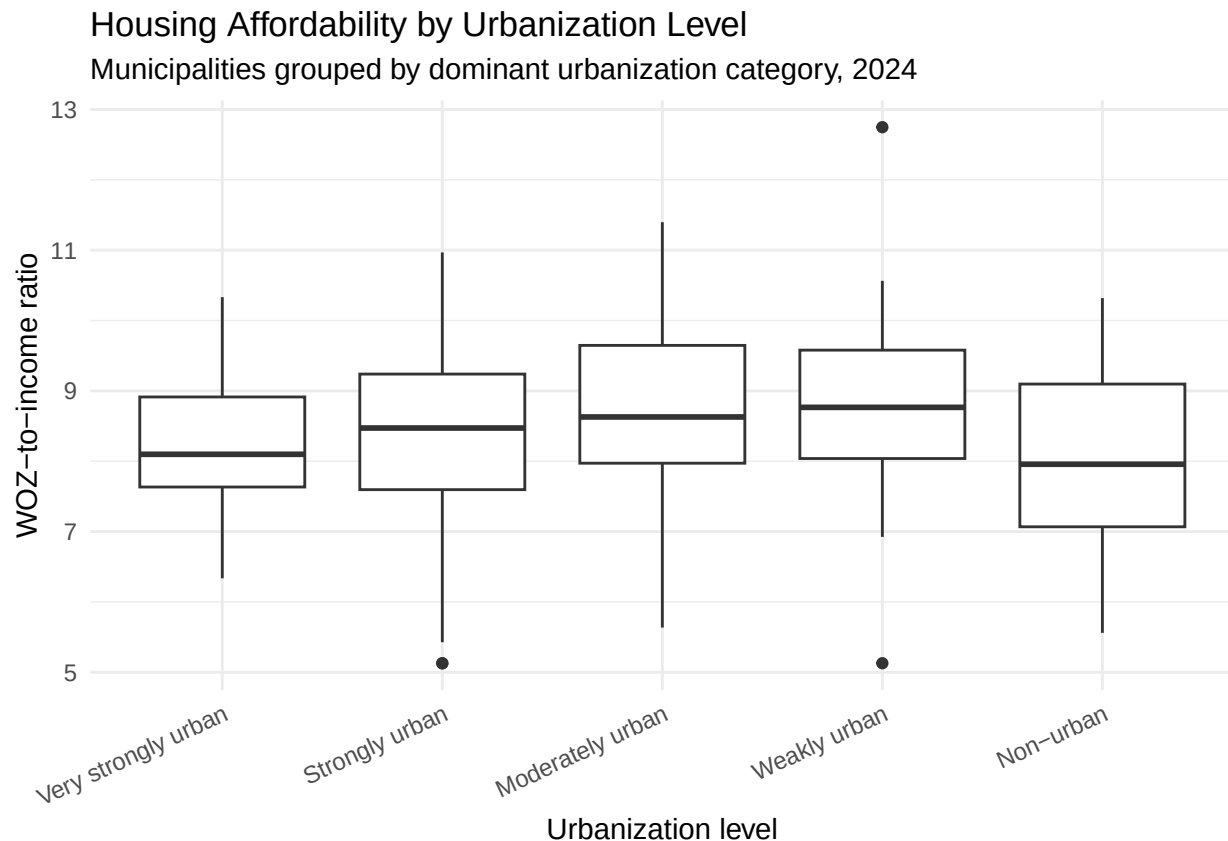
```

```

Year,
RegioS,
RegioS_label,
GemiddeldeWOZWaardeVanWoningen_98,
BronInkomenAlsWerknemer_141,
ZeerSterkStedelijk_52,
SterkStedelijk_53,
MatigStedelijk_54,
WeinigStedelijk_55,
NietStedelijk_56
) %>%
rename(
  GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
  InkomenWerknemer = BronInkomenAlsWerknemer_141
) %>%
mutate(
  AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
) %>%
pivot_longer(
  cols = c(
    ZeerSterkStedelijk_52,
    SterkStedelijk_53,
    MatigStedelijk_54,
    WeinigStedelijk_55,
    NietStedelijk_56
  ),
  names_to = "UrbanizationType",
  values_to = "UrbanizationValue"
) %>%
group_by(RegioS, RegioS_label, Year) %>%
slice_max(UrbanizationValue, n = 1, with_ties = FALSE) %>%
ungroup() %>%
mutate(
  UrbanizationType = recode(
    UrbanizationType,
    ZeerSterkStedelijk_52 = "Very strongly urban",
    SterkStedelijk_53 = "Strongly urban",
    MatigStedelijk_54 = "Moderately urban",
    WeinigStedelijk_55 = "Weakly urban",
    NietStedelijk_56 = "Non-urban"
  ),
  UrbanizationType = factor(
    UrbanizationType,
    levels = c(
      "Very strongly urban",
      "Strongly urban",
      "Moderately urban",
      "Weakly urban",
      "Non-urban"
    )
  )
)

```

```
ggplot(urban_affordability, aes(x = UrbanizationType, y = AffordabilityRatio)) +
  geom_boxplot() +
  labs(
    title = "Housing Affordability by Urbanization Level",
    subtitle = "Municipalities grouped by dominant urbanization category, 2024",
    x = "Urbanization level",
    y = "WOZ-to-income ratio"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 25, hjust = 1)
  )
```



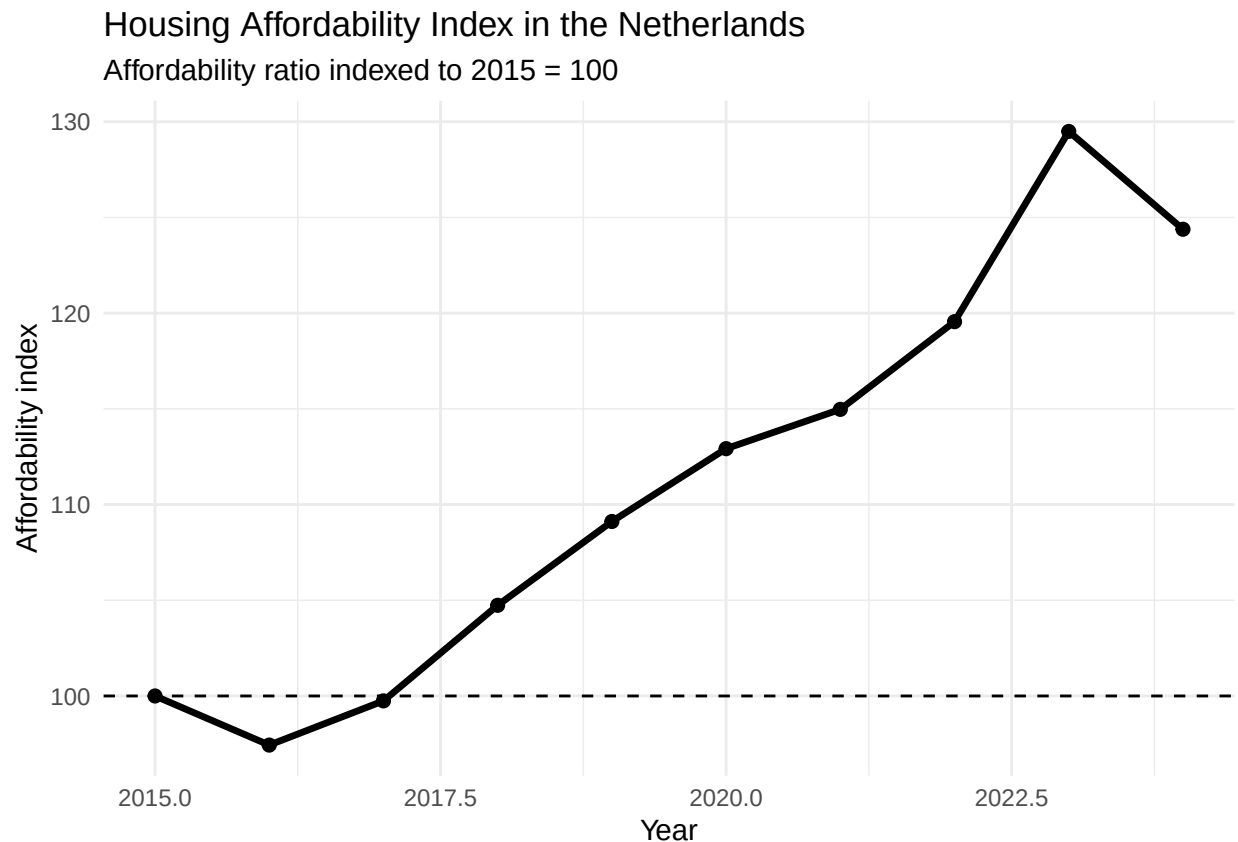
3.6 Event analysis

Analyze the relationship between two variables. Here you provide a description of why the plot above is relevant to your specific social problem.

```
landelijk_affordability_index <- landelijk %>%
  arrange(Year) %>%
  mutate(
    AffordabilityIndex = AffordabilityRatio / AffordabilityRatio[Year == 2015] * 100
  )

ggplot(landelijk_affordability_index, aes(x = Year, y = AffordabilityIndex)) +
  geom_line(linewidth = 1.2) +
```

```
geom_point(size = 2) +
geom_hline(yintercept = 100, linetype = "dashed") +
labs(
  title = "Housing Affordability Index in the Netherlands",
  subtitle = "Affordability ratio indexed to 2015 = 100",
  x = "Year",
  y = "Affordability index"
) +
theme_minimal()
```



Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.